1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

ANS:-

Given:

- Network address: 192.168.10.0/24
- Original subnet mask: 255.255.255.0
- Required number of subnets: 4

To create 4 equal subnets, we borrow 2 bits from the host portion because (2^2 = 4).

New CIDR prefix: /26

New subnet mask: 255.255.255.192

Block size: (256 - 192 = 64)

Subnet	Network address	Subnet mask	Broadcast address
1	192.168.10.0/26	255.255.255.192	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.255

Each subnet contains 64 total addresses, of which 62 are usable host addresses.

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Hint: Make sure not to use the network or broadcast address for any device.

ANS:-

Assigning IP Addresses to Five Devices

Let's use Subnet 1: 192.168.10.0/26.

- Network address: 192.168.10.0
- Usable host range: 192.168.10.1 – 192.168.10.62
- Broadcast address: 192.168.10.63
- Subnet mask: 255.255.255.192

Device	Assigned IP address	Subnet mask

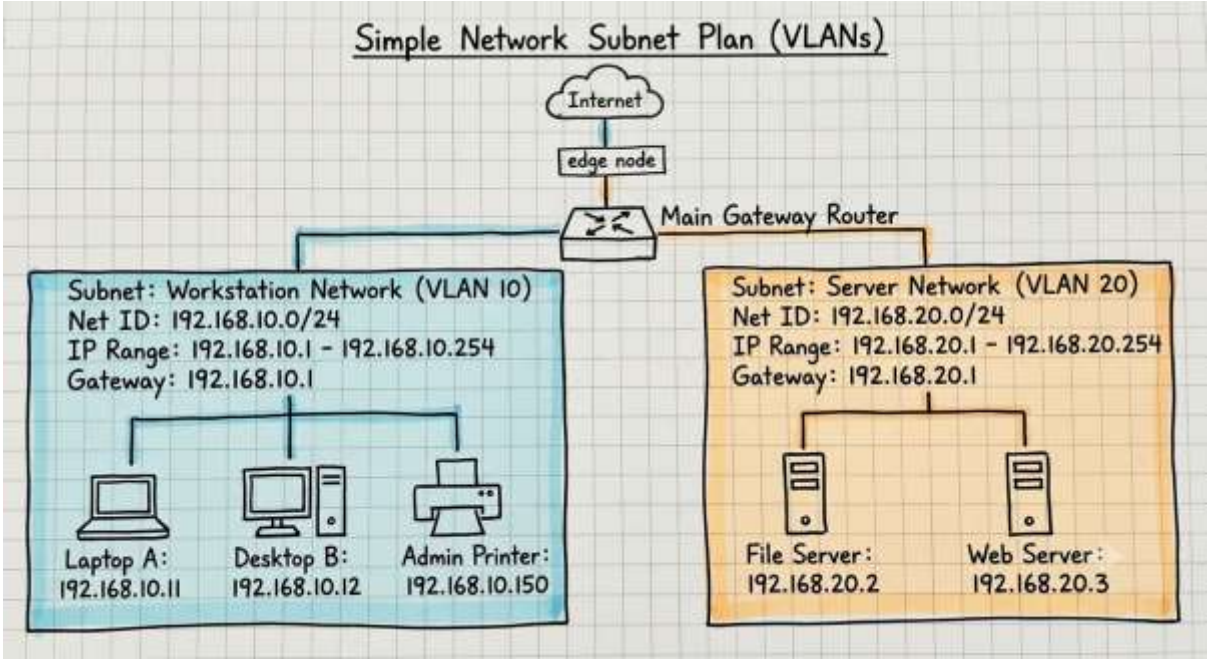
Router (gateway)	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile phone	192.168.10.5	255.255.255.192

The default gateway for all five devices is 192.168.10.1.

Explanation: All assigned addresses are unique and belong to the usable host range. Neither the network address nor the broadcast address is assigned to a device.

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

ANS:-



4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

ANS:-

Step-by-Step Subnetting Calculation

Step 1: Find the number of bits to borrow.

We need 4 subnets. Since 2^2 = 4, we borrow 2 bits from the host portion.

Step 2: Calculate the new subnet mask.

The original prefix is /24. Adding the 2 borrowed bits gives:

24+2=26

Therefore, the new prefix is /26, and the subnet mask is 255.255.255.192.

Step 3: Calculate the number of host bits.

An IPv4 address contains 32 bits.

32-26=6 host bits

Step 4: Calculate addresses per subnet and usable hosts.

2^6=64 total addresses

64-2=62 usable host addresses

We subtract 2 because each subnet reserves one address for the network and one for the broadcast.

Step 5: Calculate the block size.

The last octet of the subnet mask is 192.

256-192=64

Therefore, the subnet network addresses increase in steps of 64: 0, 64, 128, and 192.

Step 6: Find the broadcast addresses.

The broadcast address is one less than the next subnet's network address:

- Subnet 1: 192.168.10.63
- Subnet 2: 192.168.10.127
- Subnet 3: 192.168.10.191
- Subnet 4: 192.168.10.255

Step 7: Assign device IP addresses.

Choose unique IP addresses from the usable host range of the selected subnet. For Subnet 1, the router receives 192.168.10.1, and the remaining devices receive 192.168.10.2 through 192.168.10.5.

Conclusion: The 192.168.10.0/24 network can be divided into four equal /26 subnets, each supporting 62 usable hosts. Correct subnetting avoids address conflicts and allows devices to communicate within their assigned network.